Chapter 3

Stains and Staining Methods

EEGYRF

Chapter Outline

3.1 Techniques in Observing Microorganisms

3.2 Purpose of Staining

3.3 Stains

3.4 Principle of Staining

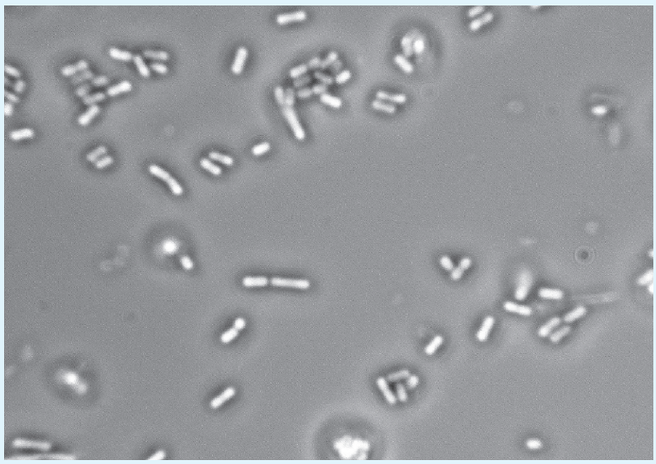
3.5 Preparation of Materials for Staining

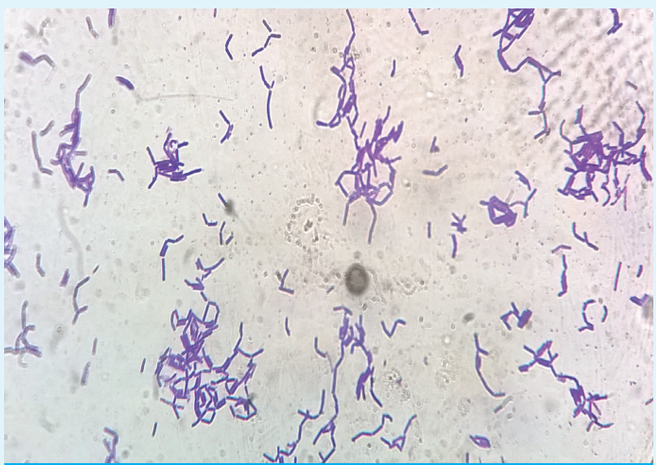
3.6 Simple Staining Method

3.7 Differential Staining

3.8 Special Staining – Endospore Staining

3.9 Commonly used Stains and its Applications





Unstained and stained *Lactobacillus sp*. in curd. *Lactobacillus* is a genus of bacteria which can convert lactose in milk into lactic acid by means of fermentation. Staining is used to visualize microbial cells under a microscope.

Learning Objectives

After studying this chapter the student will be able,

- *To appreciate the need for staining.*
- *To differentiate between an acidic dye and a basic dye and understand the principle of staining.*
- *To classify organisms based on staining reaction and differentiate between simple and differential stains.*
- *To know smear preparation and heat fixation.*
- *To describe the procedure of simple, Gram's and endospore staining methods.*
- *To describe the appearance of Gram positive and Gram negative cells after each step of Gram staining procedure.*
- *To know the importance of Gram staining and endospore staining in diagnosing and identifying bacteria.*
- *To learn a few staining solutions and names of bacteria.*